Ni-WC Dissolution: Classifying Overlay Wear Performance

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Objectives

Goal

To semi-quantitatively display the relationship between the *dissolution of carbides* and *wear resistance* of *Metal Matrix Composite* (MMC) overlays.

Methodology and Scope

- The following model **retrospectively analyzes** multiple MMC overlays' final **carbide volume fractions** ($f_{vol,WC}$)
- Applies a **mass balance** to determine **dissolution** (D) in units of **milligrams per cm of overlay**.
- Image analysis gives an overlay's a *mean free path* (MFP).
- Subsequently both **dissolution** and **MFP** can be correlated to **wear resistance**.
- Displayed here is a new method to interpret the dissolution of MMC overlays.

Background

What are metal matrix composite overlays?

MMC overlays are traditionally used in abrasive applications where high wear resistance is required. A tough matrix alloy is used with hard carbides to provide optimal wear performance.

What are practical applications?

A common application in the oil and gas industry is the hydro-transport of slurry mixtures. By cladding the inside of carbon steel pipelines with MMC overlays, such as Ni-WC, one can significantly improve the operational lifetime of the system.

What is the problem?

The concentrated heat profiles of welding processes such as GMAW, PTAW and laser cladding have tendencies to dissolve the carbides during deposition. When these carbides reduce in size and re-precipitate as weaker secondary phases, the overlay's wear resistance decreases.

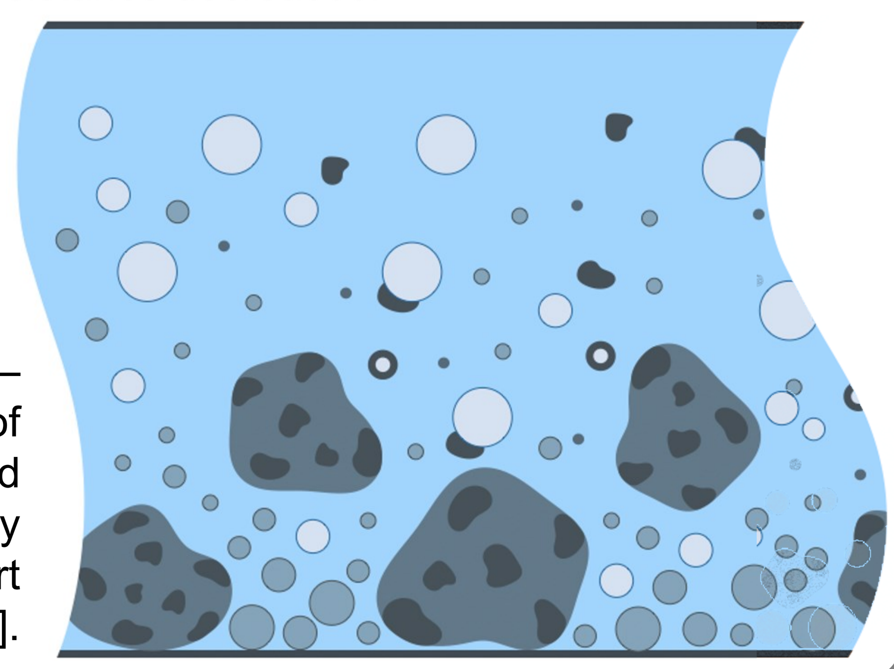

Figure 1 – Schematic of suspended particles in slurry hydrotransport [1].

Flow of Tasks

Survey published Ni-WC overlay micrographs

Clad composite material on base metal

Measure initial and final carbide **mass fractions**

$$f_{mass,WC} = \frac{f_{vol,WC} \cdot \rho_{WC}}{\rho_{mixture}}$$

Convert mass fraction to **linear mass (m'_{WC}) [mg/cm]**

$$m'_{WC,initial} = f_{mass,WC} \cdot A_{WC} \cdot \rho_{WC}$$

$$m'_{WC,overlay} = \frac{f_{mass,WC} \cdot \dot{m}_{WC}}{V_{travel}}$$

Calculate **dissolution ($m'_{WC,diss}$)**

$$D = \eta_{transfer} m'_{WC,initial} - m'_{WC,overlay}$$

Measure mean free path (MFP) of carbides

$$\mathrm{MFP} = \frac{1}{N}\sum_{j=1}^{N}\left(\frac{1}{\mathrm{CN}}\sum_{i=1}^{\mathrm{CN}} d_i\right)_j$$

Correlate to **ASTM G65** wear test

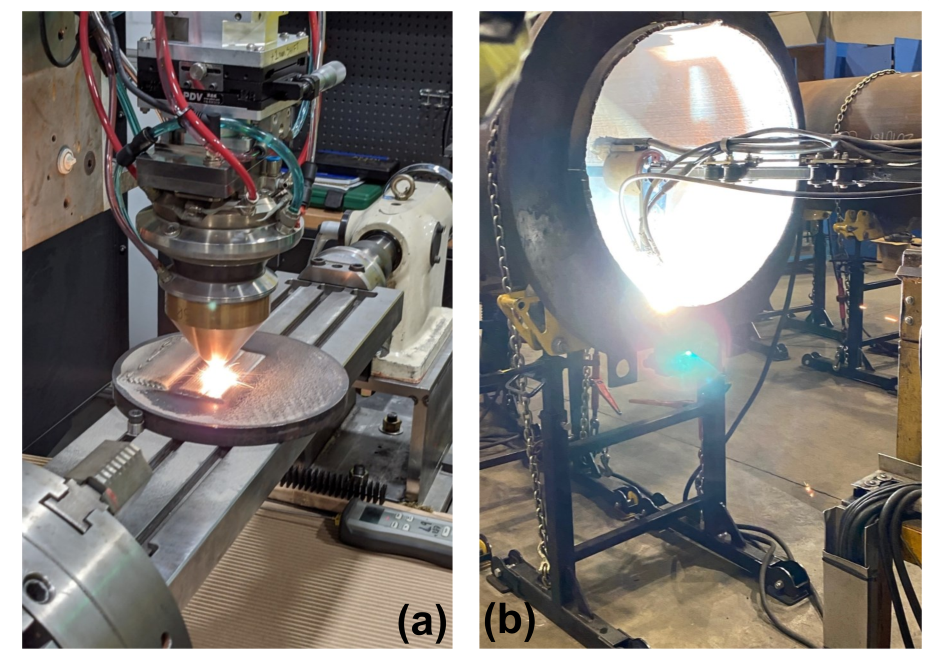

Figure 2 – Bead-on-plate laser cladding (a) and internal pipe cladding via GMAW (b).

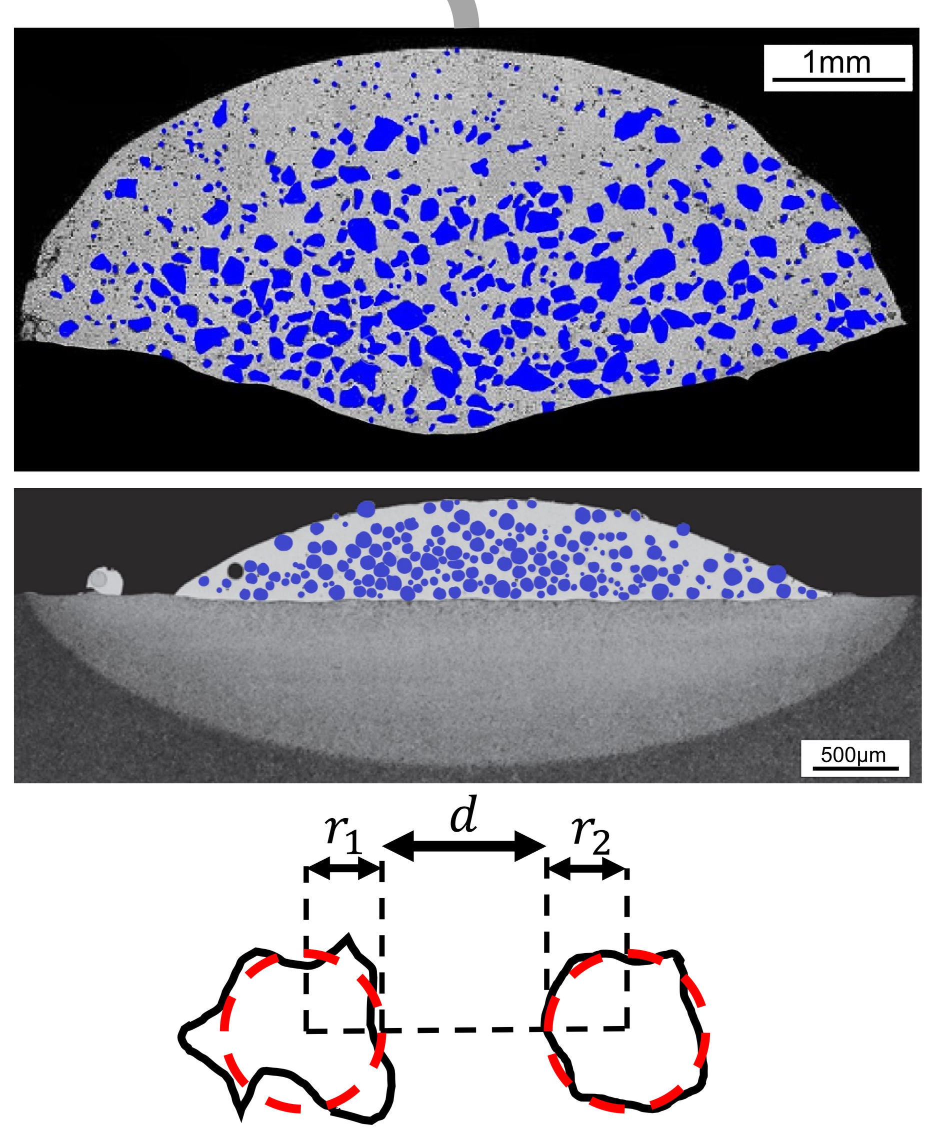

Figure 3 – Carbide colored images of an (a) STT-GMAW process [2] and a (b) laser process [3], along with a (c) schematic of determining carbide mean free path.

Justification of procedure

- This method uses a conceptual transfer efficiency ($\eta_{transfer}$) to acknowledge the effects of carbide "non-wetting" as documented by Guest et al [4] (see *Raw Data* QR Code).
- To account for **secondary re-reprecipitated carbides**, this method uses a low-pass threshold size of 30μm
  - Approximate wire fed process mesh size: 100-180μm
  - Approximate powder fed process mesh size: 50-100μm

Results

- Plotting a **fitted exponential trend** of **MFP vs G65 Mass Loss** (i.e. *G(MFP)*) to **published data [5]** with the measured semi-quantitative relationship of **MFP vs D** yielded the following.
- Declaring a $\mathrm{MFP}_{critical}$ of ~106μm extrapolates a $\mathrm{D}_{critical} \approx 230$ mg/cm

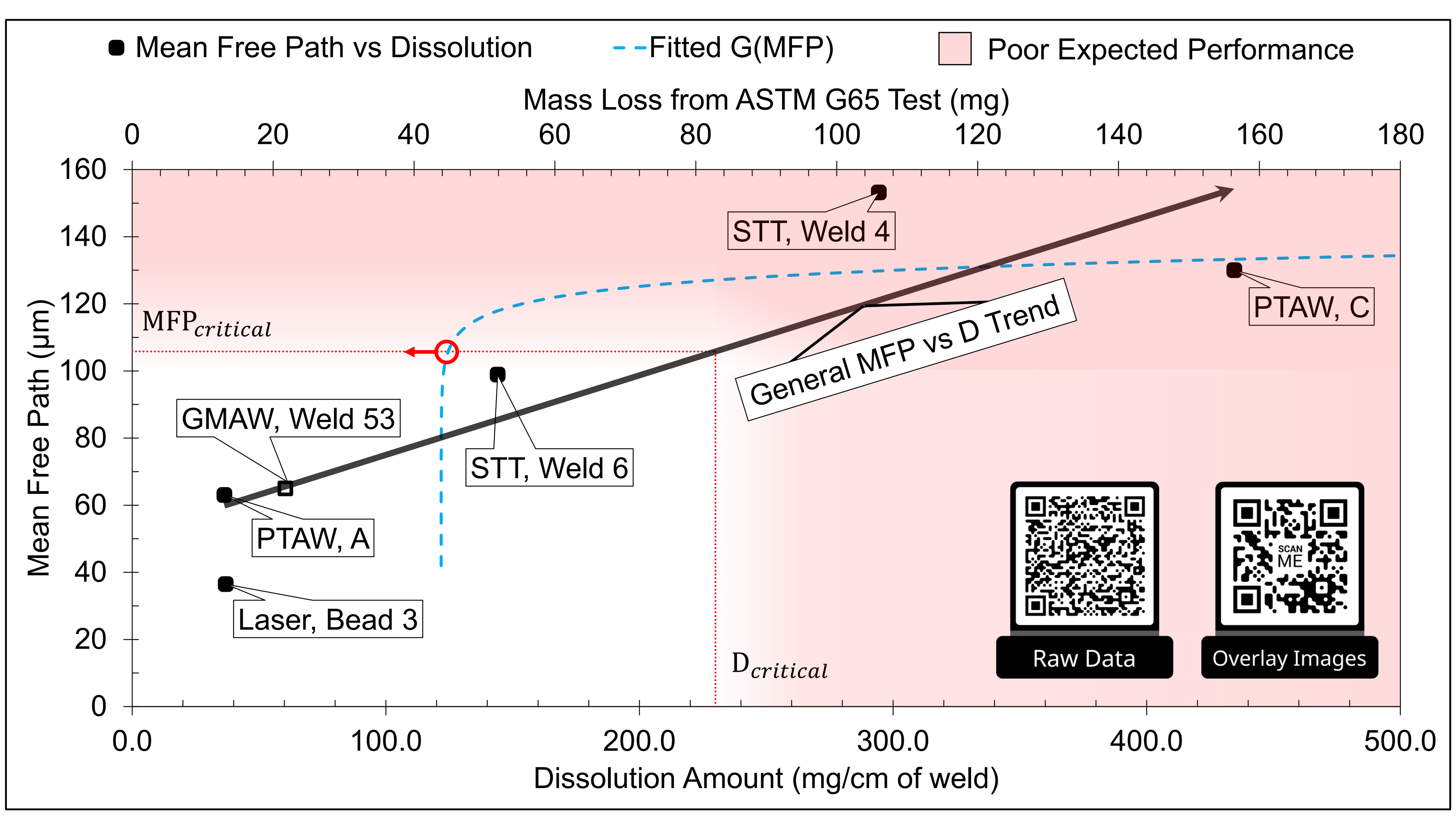

Figure 4 – The modelled relationship of MFP vs D and how it interplays with wear resistance (i.e. mass loss).

Discussion

General principles from the model

- **Only when both MFP < $\mathrm{MFP}_{critical}$ and D < $\mathrm{D}_{critical}$ can an overlay be expected to perform well**
- **Critical values** are not discrete, but rather **transition** regions
- All **solid points successfully corroborate** with visual inspection of overlays
- **MFP vs D**'s relationship **cannot** be expected to be **linear**; however, it does demonstrate an **increasing relationship**

Errors and uncertainty

- **Reliance** on the **accuracy** of published process parameters
- **ImageJ** can be **inaccurate** for poorly coloured images
- **MFP vs G65 mass loss has yet to be quantified**

Additional assumptions

- **Area** fraction of carbides **directly infers volume** fraction
- Carbide MFP can be approximated with a **coordination number** ($\mathrm{CN} = 4$)
- Dissolution has negligible effects on material densities
- $\mathbf{MFP}_{critical}$ can be **approximated** by solving the following:

$$20\log(\mathrm{G}(\mathrm{MFP}_{critical})) = 0$$

Main limitations

- There is **limited applicable data/micrographs**
- **MFP's effect** on **wear rate** is **dependent** on the abrasive **test** used [6]

Conclusions

Real world applicability

- The above model is new concept that can **aid** in the **classification** of **metal matrix composite overlays**.
- When an overlay is plotted as an MFP vs D point, the user **instantly** gains semi-quantitative **insight** on the wear **performance** of that overlay **without** ASTM G65 **testing**.

What to do next?

- **Quantitatively** cross reference the ranked performance of overlays via this model and via ASTM G65 testing.

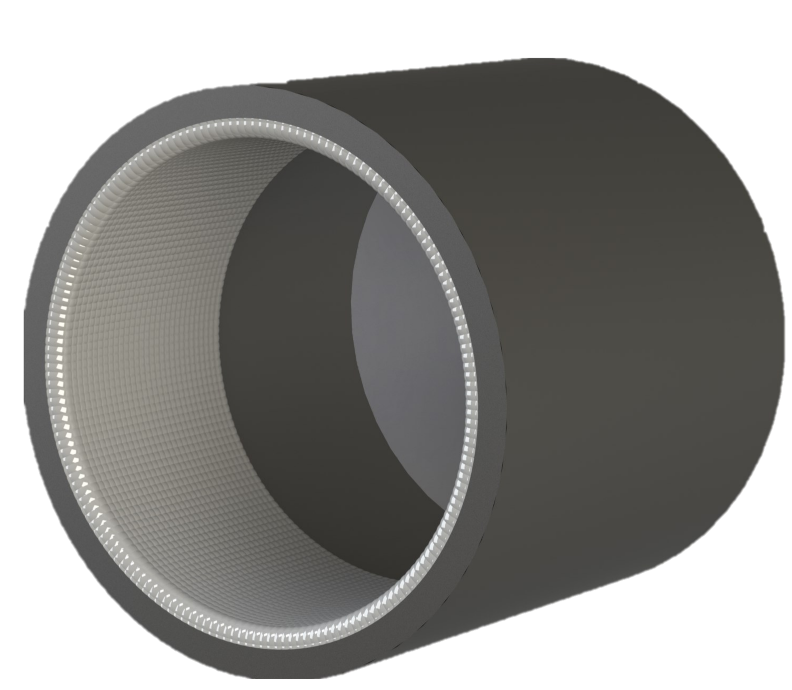

Figure 5 – CAD render of an internally cladded pipeline used for slurry hydro-transport.

Acknowledgments

Cited Works

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